Practice Problem Set 1

ECON 320 – Introduction to Mathematical Economics

Fall 2025

Instructions. Show all essential steps. Unless specified, provide exact answers (no decimals). Clearly state units and interpret results when an economic context is given.

1. Solve each inequality for x and express the solution in interval notation.
   (a) $|2x-3| \le 5$.
   (b) $(x-3)(x+2) \le 0$.
   (c) $\dfrac{1}{x-2} \ge 3$.
2. Let $A = [-1, 4)$ and $B = (1, 6]$. Find:
   (a) $A \cup B$ and $A \cap B$,
   (b) $A \setminus B$ and $B \setminus A$,
   (c) $\inf(A \cap B)$, $\sup(A \cap B)$.
3. Determine the *domain* of
$$f(x) = \frac{\sqrt{5 - \ln(x^2 - 9)}}{x-3}.$$
   Write the domain as a union of intervals and justify briefly.
4. Consider $f(x) = \dfrac{2x-3}{x+1}$ with domain $x > -1$, $x \neq -1$.
   (a) Find the inverse function $f^{-1}(y)$ and state its domain.
   (b) Evaluate $f^{-1}(5)$.
5. Exponential–logarithmic equations. Solve for x (exact form preferred):
   (a) $e^{3x} = 7$.
   (b) $2^{x+1} = 5e^{0.2x}$ (give x to three decimal places).

6. Sequences and bounds. Let $S = \left\{ \frac{2n+1}{n+3} : n \in \mathbb{N} \right\}$.

   (a) Compute $\lim_{n\to\infty} \frac{2n+1}{n+3}$.

   (b) Find $\inf S$ and $\sup S$ (justify).

7. Sets in $\mathbb{R}^2$. Let $B = \{(x,y) \in \mathbb{R}^2_{\geq 0} : 2x + 3y \leq 60\}$.

   (a) Find the x- and y-intercepts of the boundary $2x + 3y = 60$.

   (b) Compute the area of B.

   (c) Write B as $\{(x,y) : 0 \leq x \leq \bar{x},\ 0 \leq y \leq g(x)\}$ by solving for y in terms of x.

8. Composition and transformation. Let $u(x) = \ln x$ (domain $x > 0$) and $g(x) = 2 - 3\ln(4 - x)$.

   (a) State the domain of g.

   (b) Write g as an affine transformation of $u(\cdot)$ composed with a linear function. Identify the horizontal shift and reflection.

9. Absolute value with parameters. For $a \in \mathbb{R}$ and $b > 0$, solve

$$|x - a| \leq b,$$

   and express the solution set as an interval in terms of a and b.

10. Domains with radicals and rationals. Determine the domain of

$$h(x) = \sqrt{9 - x^2} + \frac{1}{\sqrt{x-1}}.$$

   Write the final domain as a (possibly empty) interval or union of intervals.

11. Differentiate and simplify. Then evaluate at the indicated point.

   (a) $y = \frac{(3x^2+1)^5}{x\,e^{2x}}$, find $y'(1)$.

   (b) $y = \ln\left(\sqrt{1+4x^2}\right)$, find y' and y''.

12. Cost and average cost. Let $C(q) = 0.5q^2 + 10q + 400$ with $q \geq 0$.

   (a) Find marginal cost $MC(q)$ and average cost $AC(q)$.

   (b) Compute $MC(20)$ and $AC(20)$.

(c) Find the output $\hat{q}$ that minimizes $AC(q)$ and the minimum average cost.

13. Elasticity of demand.

    (a) For $Q(P) = 1000P^{-1.2}$, compute the price elasticity $\varepsilon(P)$ and state whether it depends on P.

    (b) For $Q(P) = 200 - 2P$, compute ε at $P = 50$.

14. Implicit differentiation. For the curve

$$xe^x + y^2 = 10,$$

find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ as a function of (x, y) and then evaluate at the point with $x = 1$ and $y > 0$.

15. Total differential and proportional changes. Suppose output is

$$Q = K^{0.3}L^{0.7}P^{0.1}.$$

At the base point $(K, L, P) = (100, 400, 2)$, approximate the percentage change in Q if K increases by 5%, L decreases by 2%, and P increases by 10% (use differentials).

16. Linear supply–demand with a per-unit tax t. Let

$$Q_s = -20 + 4(P - t), \qquad Q_d = 500 - 6P + 2Y,$$

with $Y = 100$.

    (a) Solve for the equilibrium price $P(t)$ and quantity $Q(t)$ as functions of t.

    (b) Compute $\dfrac{\mathrm{d}P}{\mathrm{d}t}$ and $\dfrac{\mathrm{d}Q}{\mathrm{d}t}$ and evaluate at $t = 5$.

    (c) Briefly interpret $\dfrac{\mathrm{d}P}{\mathrm{d}t}$ in words (tax incidence on buyers).

17. Profit maximization and comparative statics. Let $\pi(q) = pq - (aq + \frac{b}{2}q^2)$ with parameters $p > 0$, $b > 0$.

    (a) Find the profit-maximizing $q^*(p, a, b)$.

    (b) Compute $\frac{\partial q^*}{\partial p}$ and $\frac{\partial q^*}{\partial a}$ and sign them.

18. Elasticity of supply. If $Q(P) = cP^\eta$ with $c > 0$, $\eta > 0$,

    (a) Derive the price elasticity of supply $\varepsilon_s(P)$.

    (b) For $c = 5$ and $\eta = 0.8$, compute ε_s at $P = 4$ and $P = 25$ and comment on constancy.

19. Implicit-function comparative statics. Suppose $F(x,a) = x^3 - 4x + a = 0$ defines $x = x(a)$ near a root.

   (a) Use the implicit-function rule to find $\dfrac{\mathrm{d}x}{\mathrm{d}a}$ in terms of x.

   (b) Evaluate $\dfrac{\mathrm{d}x}{\mathrm{d}a}$ at a root $x = 2$.

20. Multivariate total derivative (dynamic comparative statics). Let market equilibrium be defined by

$$\begin{cases} D(P,Y) = \alpha Y - \beta P = Q, \\ S(P,W) = \gamma P - \delta W = Q, \end{cases}$$

with positive parameters. Treat $Y = Y(t)$ and $W = W(t)$ as time-varying exogenous variables.

   (a) Solve for P and Q as functions of Y and W.

   (b) Compute $\frac{\mathrm{d}P}{\mathrm{d}t}$ and $\frac{\mathrm{d}Q}{\mathrm{d}t}$ in terms of $\dot{Y}$ and $\dot{W}$.

   (c) Evaluate $\frac{\mathrm{d}Q}{\mathrm{d}t}$ when $\alpha = \beta = \gamma = \delta = 2$, $Y = 100$, $W = 60$, $\dot{Y} = 3$, $\dot{W} = -1$.

21. Log-linear approximation. Let $R(q) = Aq^{\theta}$ with $A > 0$, $\theta \in (0,1)$ and cost $C(q) = c_0 + c_1 q$ ($c_0, c_1 > 0$). Profit $\pi(q) = R(q) - C(q)$.

   (a) Compute $MR(q)$ and $MC(q)$.

   (b) Using a first-order (log) approximation around $q = q_0$, derive an approximate expression for the change in profit $\Delta\pi$ from a small percentage change in q.

   (c) At $A = 200$, $\theta = 0.6$, $c_0 = 100$, $c_1 = 20$, $q_0 = 50$, estimate $\Delta\pi$ for a $+2\%$ change in q.

22. Comparative statics with a subsidy. Demand: $Q_d = 300 - 3P$. Supply with per-unit subsidy $s > 0$: $Q_s = -30 + 2(P + s)$.

   (a) Solve for the equilibrium $P(s)$ and $Q(s)$.

   (b) Compute $\dfrac{\mathrm{d}P}{\mathrm{d}s}$ and $\dfrac{\mathrm{d}Q}{\mathrm{d}s}$ and interpret the signs.

23. Implicit elasticity. Suppose demand satisfies $F(Q,P) = \ln Q + \phi Q - \ln P = 0$ with $\phi > 0$ (defining $Q = Q(P)$).

   (a) Using implicit differentiation, find $\dfrac{\mathrm{d}Q}{\mathrm{d}P}$.

   (b) Derive the price elasticity $\varepsilon = \dfrac{\mathrm{d}Q}{\mathrm{d}P} \cdot \dfrac{P}{Q}$ in terms of Q (or P).